$R=\{((a, b),(c, d)) | ad = bc\}$ where $W = \mathbb{Z}\times\mathbb{N}$

1. Reflexive: To prove if this relation is reflexive we test
   $\forall x, x \in W \rightarrow (x, x) \in R$
   Because ad = bc it follows that xx = xx and that (x,x) ∈ R
   ∴ This relation is reflexive. ■

2. Symmetric: To prove if this relation is symmetric we test

   $\forall x, \forall y, (x, y) \in R \rightarrow (y, x) \in R$

   Because ad = bc from (a, b),(c, d) that would make (c, d), (a,b) which would become db = ac which is not equivalent to the term ad = bc with a, b, c, d in relation W.

   ∴ This relation is not symmetric. ■

3. Transitive: To prove if this relation is transitive we test

   $\forall x\ \forall y\ \forall z\ [(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R]$

   Because ad = bc that would mean that for another term, z, in this case, (e, f), with (c, d), it would equate to cf = de. So if (a, b), (c, d) ∈ R and (c, d), (e, f) ∈ R it would imply that (a, b), (e ,f) ∈ R which would become af = eb.

   However, when you simplify for c and d from cf = de you get c = $\frac{de}{f}$ and d = $\frac{cf}{f}$ and plug these values for c and d back into ab = bc you get a result of ac = bd which is not equivalent, then (a, b), (e ,f) ∉ R.

   ∴ This relation not transitive. ■